Operating Systems
Week 9
Lecture/Discussion 9

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Previous

- Deadlock avoidance algorithms

Today Discussion

- Continue deadlock avoidance algorithms

$$Available = Available - Request_i;$$
$$Allocation_i = Allocation_i + Request_i;$$
$$Need_i = Need_i - Request_i;$$

Example: P_1 Request (1,0,2)

Check that Request ≤ Available (that is, (1,0,2) ≤ (3,3,2) ⇒ true

	Allocation	*Need*	*Available*
	A B C	*A B C*	*A B C*
T_0	0 1 0	7 4 3	2 3 0
T_1	3 0 2	0 2 0	
T_2	3 0 2	6 0 0	
T_3	2 1 1	0 1 1	
T_4	0 0 2	4 3 1	

Executing safety algorithm shows that sequence < T_1, T_3, T_4, T_0, T_2> satisfies safety requirement

Need is defined to be ***Max - Allocation***

Can request for (3,3,0) by T_4 be granted?

Can request for (0,2,0) by T_0 be granted?

Example: P_1 Request (1,0,2)

- **T1**: Need = [0, 2, 0] ☑ → Run T1 → Release [3, 0, 2] → Work = [5, 3, 2]
- **T3**: Need = [0, 1, 1] ☑ → Run T3 → Release [2, 1, 1] → Work = [7, 4, 3]
- **T4**: Need = [4, 3, 1] ☑ → Run T4 → Release [0, 0, 2] → Work = [7, 4, 5]
- **T0**: Need = [7, 4, 3] ☑ → Run T0 → Release [0, 1, 0] → Work = [7, 5, 5]
- **T2**: Need = [6, 0, 0] ☑ → Run T2 → Release [3, 0, 2] → Work = [10, 5, 7]

- Safe Séquence = [T1, T3, T4, T0, T2]

Algorithm Cont..

Current Available = [3, 3, 2]

Can request for (3,3,0) by T4 be granted?

Algorithm Cont..

(From initial system state before any new request)

T4 Details

T4 Max = [4, 3, 3]

T4 Allocation = [0, 0, 2]

T4 Need = Max - Allocation = [4, 3, 3] - [0, 0, 2] = [4, 3, 1]

- T4 Request = [3, 3, 0]
- Is request ≤ Available?

  Request = [3, 3, 0]
  Available = [3, 3, 2]
  [3 ≤ 3, 3 ≤ 3, 0 ≤ 2]

  Yes, So, **the request is valid** and can be **provisionally granted**.

Algorithm Contd…

🛠️ Step 3: Simulate Provisional Allocation
🔄 New State After Granting Request:
New Available = [3, 3, 2] - [3, 3, 0] = [0, 0, 2]
New T4 Allocation = [0, 0, 2] + [3, 3, 0] = [3, 3, 2]
New T4 Need = [4, 3, 1] - [3, 3, 0] = [1, 0, 1]

Algorithm Contd…

Step 4: Check for Safe State
Use updated system state:
Updated Available = [0, 0, 2]
Updated Allocation Matrix:

Updated Allocation Matrix
T0 = [0, 1, 0]
T1 = [2, 0, 0]
T2 = [3, 0, 2]
T3 = [2, 1, 1]
T4 = [3, 3, 2] ← updated

Algorithm Contd…

Max Matrix (unchanged):
T0 = [7, 5, 3]
T1 = [3, 2, 2]
T2 = [9, 0, 2]
T3 = [2, 2, 2]
T4 = [4, 3, 3]

Algorithm Contd…

Recalculate Need = Max - Allocation:
T0 = [7, 4, 3]
T1 = [1, 2, 2]
T2 = [6, 0, 0]
T3 = [0, 1, 1]
T4 = [1, 0, 1] ← updated